

ICT Infrastructure and System Support Assessment Report

Organization: Yared Tour & Travel

Prepared for: Management

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Date: September 2026

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Executive Summary

This report presents a comprehensive assessment of the Information and Communication Technology (ICT) infrastructure and system support mechanisms at Yared Tour & Travel. The assessment was conducted through management and staff interviews, physical inspection of hardware assets, software review, network performance testing, and evaluation of the organization's digital presence (website and social media).

The findings reveal an organization is fairly functioning but some critical issues which require critical attention. While Yared Tour & Travel has built a strong brand identity rooted in responsible, community-based tourism since 2004, its technology should be transformed with the growing demands to stay and expand in the sector. Therefore, in light of the available resources and management's commitments, the organization's digital systems and online presence should be revised in accordance with the findings and proposed recommendations.

1. Introduction and Background of Tour and Travel Organizations

1.1 The Nature of Tour and Travel Operations

Tour and travel organizations operate at the intersection of information, logistics, and human experience. A tour operator's core product is not a tangible good but an orchestrated sequence of experiences—transportation, accommodation, guided activities, cultural encounters, and logistical coordination—delivered across multiple locations and often involving numerous third-party suppliers.

This operational model makes tour and travel businesses uniquely dependent on timely, accurate information flow. A single booking involves coordination between the traveler, the tour operator, local guides, accommodation providers, transport services, and often international partners. Each of these touchpoints generates data that must be captured, processed, and communicated reliably.

1.2 The Digital Transformation Imperative

The global tourism industry has undergone profound digital transformation over the past decade. Research confirms that efficiency, productivity, profitability, effectiveness, and marketing are all improved after employing new technologies, leading to the conclusion that the integration of technology in tourism is unavoidable for the continued existence of service providers in the market.

This transformation is not merely about adopting new tools—it represents a fundamental shift in how tourism businesses operate, compete, and deliver value. Travelers now expect:

- Instant access to information across multiple digital channels
- Seamless online booking and payment capabilities
- Personalized experiences informed by data and artificial intelligence
- Real-time communication before, during, and after their journey

1.3 Yared Tour & Travel: Organizational Profile

Yared Tour & Travel is an Ethiopian-Dutch tour operator that has been designing immersive, sustainable journeys across Ethiopia since 2004, with a European office in the Netherlands supporting its international clientele. The organization specializes in:

- Community-based and cultural tours across Ethiopia, Djibouti, Kenya, and Rwanda
- Tailor-made itineraries for individuals and small groups
- Responsible, sustainable travel that generates income opportunities in rural communities

The organization's mission is to "develop responsible community tourism experiences" while showcasing Ethiopia's culture, nature, and history through ethical and respectful travel. As a Travelife Partner since 2019, Yared Tour & Travel has demonstrated a commitment to sustainability and responsible tourism practices.

1.4 The Strategic Role of ICT in Achieving Organizational Goals

ICT is not a peripheral administrative function—it is the operational backbone that enables the organization to:

1. Reach international markets through a professional, functional website and social media presence
2. Manage complex logistics across remote Ethiopian destinations and multiple international partners
3. Communicate reliably with clients, local guides, and accommodation providers
4. Process financial transactions securely and maintain accurate records for tax compliance
5. Showcase the organization's unique value proposition through compelling digital content

This report assesses the current state of ICT infrastructure and system support within Yared Tour & Travel, identifies gaps and limitations, and provides actionable recommendations for improvement.

2. ICT and Tour and Travel

2.1 The Role of ICT in Tourism Operations

Information and Communication Technology (ICT) plays a multifaceted role in modern tourism operations. According to industry research, ICTs allow supply chain management and customer management relations to be combined into a single source that facilitates product selection, fulfillment, ordering, payment, tracking, and reporting through integrated systems. Management within tourism companies uses ICTs to undertake a range of tasks that enhance the efficiency of employees, particularly in online reservations and customer communications.

The role of ICT in tourism can be categorized into five key domains:

Domain	Function	Impact on Yared Tour & Travel
Customer-Facing Technology	Website, social media, booking systems, communication tools	Enables international market reach and direct client engagement
Operational Systems	Booking management, itinerary planning, supplier coordination	Supports complex logistics across remote destinations
Financial Systems	Invoicing, payment processing, accounting, tax compliance	Ensures regulatory compliance and financial transparency
Communication Infrastructure	Email, messaging, video conferencing, VoIP	Connects Ethiopian and Netherlands offices with clients and partners

Data & Analytics	CRM, marketing analytics, performance tracking	Informs strategic decisions and marketing investments
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2.2 The Information-Intensive Nature of Tourism

Tourism is fundamentally an information-driven industry. As researchers have noted, "Information is the key element in the tourism industry". Travelers depend on accurate, timely information to make decisions about destinations, accommodations, activities, and logistics. Tour operators, in turn, depend on information systems to manage the complex web of relationships and transactions that constitute a travel experience.

For a community-based tour operator like Yared Tour & Travel, this information dependency is even more pronounced. Operating across remote Ethiopian destinations requires:

- Reliable communication channels to coordinate with local guides and community hosts
- Robust booking systems to manage reservations across multiple locations
- Effective digital marketing to reach travelers in Europe, North America, and other international markets
- Secure financial systems to process payments and maintain compliance with Ethiopian and Dutch tax regulations

The tourism industry is currently experiencing several transformative technology trends that are relevant to Yared Tour & Travel's too to enhance the strategic planning including:

- **Artificial Intelligence (AI) and Personalization:** AI is increasingly used to create personalized travel recommendations, automate customer service through chatbots, and optimize pricing and inventory management.
- **Data-Driven Decision Making:** Tourism organizations are increasingly leveraging data analytics to understand visitor behavior, optimize marketing investments, and improve operational efficiency.
- **Mobile-First Traveler Experience:** Travelers increasingly research, book, and manage their trips through mobile devices. This creates both opportunities (direct engagement) and challenges (the need for mobile-optimized websites and communication channels).
- **Sustainable Tourism and Digital Transparency:** As sustainability becomes a central concern for travelers, digital platforms are increasingly used to communicate and verify responsible tourism practices. Yared Tour & Travel's Travelife Partnership positions it well to leverage this trend—if the organization's digital presence effectively communicates its sustainability credentials.

3. Existing ICT System Assessment

3.1 Assessment Methodology

The assessment of Yared Tour & Travel's existing ICT system was conducted through:

- Interviews with management and operational staff regarding daily technology use and pain points

- Physical inspection of hardware assets at the Addis Ababa office
- Review of software installations and licensing status
- Network performance testing during business hours
- Website and online presence review

3.2 Hardware Assessment

Physical inspection of the organization's hardware assets revealed the following:

Asset Category	Quantity/ Status	Assessment Findings
Laptops (AMD)	3/ Deployed	Performance adequate for office tasks but updates, activation and security features required.
Desktops ((Office Admin, Finance/Tax)	3, but one is Non-functional	The PC designated for admin, finance and tax operations is currently working. But, one is a critical failure that is not working at all.
Tablets	Not available	In the future plan
Multi channel TV	Not Available	In the future plan
Printers	1 Deployed	Not functional
All-in-One (Printer/Scan/ Copy)	2 Deployed	Functional but showing signs of wear. Scanning function is slow, impacting administrative workflow.
CCTV / Internet Camera	Not available	For future expected to be deployed for office and site communication
External HD	Not available	It is critical to avail soon

3.3 Software Assessment

Main software that are used in the office operations are listed below

Software Category	Current State	Assessment Findings
Operating System	Windows 10/11	Windows OS is still in good status but requires activations and updates.
Office Applications	Mixed (MS Office, WPS, Google)	Still working but some updates, activations and standardization is required.
Browsers	Edge, Chrome, Mozilla	Still working, in some PCs outdated versions, posing security risks.

Finance Software	Outdated Peachtree	May not be fully compatible with newer operating systems. Likely contributing to the Finance PC failure.
Security Software	Unverified	The Microsoft (defender) security tools are activated but some PCs use 3rd party antivirus, not advisable.

3.4 Network Assessment

Network Component	Current State	Assessment Findings
Wi-Fi	Functional	Intermittent speed drops reported, especially during video calls and large file transfers.
Internet Speed	Unverified	Bandwidth requirements need assessment against current usage patterns (cloud apps, video conferencing).
Printer Sharing	Wi-Fi based	Inconsistent and error-prone. A dedicated network print server is recommended.
Network Security	Flat network	No network segmentation or firewall configuration for security.

3.5 Services Assessment

Service	Current State	Assessment Findings
Website	Live (yaredtour.com)	Branding is strong and mission-aligned. However, critical functionality gaps exist (see Section 3.6).
Finance Apps	Outdated	Critical but aging. Likely contributing to operational failures.
Social Media	Established but inconsistent	Accounts exist but posting is irregular. Significant untapped potential for engagement and lead generation.
Email	Functional	Basic email functionality. No evidence of professional email management or security protocols.

3.6 Website Assessment (Summary)

A detailed assessment of the organization's website (yaredtour.com) identified the following:
Strengths:

- Clear and compelling branding focused on responsible, community-based tourism
- Strong value proposition ("Why Travel with Yared?")
- Credibility signals (Travelife Partnership, detailed "About Us" content)
- Authentic visual content and mission-aligned messaging

Critical Limitations:

- No bookable tours or detailed itineraries — the most significant barrier to conversion
- No online booking or inquiry system — forces potential customers to leave the site
- Broken navigation — the "Tours" page returns an error
- Missing essential pages — no Terms & Conditions, Privacy Policy, or FAQ
- No blog or content marketing — limiting SEO and expertise demonstration
- No social proof integration — TripAdvisor reviews not showcased on the site

3.7 Social Media Assessment (Summary)

Platform	Current State	Assessment Findings
Instagram	Underutilized	Primary platform for travel inspiration. Requires consistent posting (3-5x/week) and Reels production.
Facebook	Underutilized	Strong for trust-building and community engagement. Requires regular posting and group creation.
YouTube	Minimal/None	Long-form video is experiencing a comeback. Untapped opportunity for destination storytelling.
TikTok	Not active	Optional but useful for younger audience reach if resources permit.
LinkedIn	Minimal	Relevant for B2B partnerships and trade relationships.

3.8 Summary of Critical Issues

The assessment reveals the following critical issues requiring immediate attention:

1. Non-functional PCs and printer — Actively halting essential business operations
2. Operating System upgrade and updates — Security compliance deadline approaching (October 2025)
3. No bookable tours on website — Primary barrier to online conversion
4. No formal IT support structure — Issues addressed reactively, leading to prolonged downtime
5. Software inconsistency — Compatibility problems and security vulnerabilities
6. Inconsistent social media presence — Missed opportunity for brand awareness and lead generation

4. Role and Responsibility of Effective ICT Management and Administration

4.1 The Strategic Importance of ICT Management

Effective ICT management is not merely a technical function—it is a strategic imperative for tour and travel organizations. As the tourism industry becomes increasingly digital, the quality of an organization's ICT infrastructure directly influences its ability to compete, grow, and deliver exceptional customer experiences.

For Yared Tour & Travel, effective ICT management must address three fundamental requirements:

1. **Operational Reliability** — ensuring that critical systems (customer-based planning, finance, booking, communication) function consistently and securely
2. **Strategic Enablement** — leveraging technology to reach new markets, improve efficiency, and enhance customer experience
3. **Risk Mitigation** — protecting organizational data, ensuring regulatory compliance, and maintaining business continuity

4.2 Core Responsibilities of ICT Management

Drawing on established frameworks for ICT management in tourism organizations, the following responsibilities are essential:

4.2.1 Infrastructure Management

Responsibility	Description	Priority
Hardware Lifecycle Management	Plan, procure, deploy, and retire hardware assets in a systematic manner	High
Network Administration	Maintain reliable internet connectivity, Wi-Fi, and network security	Critical
Software Management	Standardize software platforms, manage licenses, and ensure updates	High
Data Backup & Recovery	Implement and verify backup systems to prevent data loss	Critical

4.2.2 User Support and Training

Responsibility	Description	Priority
Help Desk / On-Demand Support	Provide timely resolution of user issues (remote and on-site)	Critical

Training & Onboarding	Ensure staff can effectively use organizational technology	High
Cybersecurity Awareness	Educate staff on phishing, password security, and data protection	High

4.2.3 Strategic Planning

Responsibility	Description	Priority
ICT Budgeting	Plan and optimize ICT investments aligned with business goals	High
Technology Roadmap	Develop a phased plan for infrastructure upgrades and digital transformation	Medium
Vendor Management	Manage relationships with ISPs, hardware vendors, and service providers	Medium

4.2.4 Digital Presence Management

Responsibility	Description	Priority
Website Administration	Ensure website functionality, security, and content updates	High
Social Media Support	Provide technical support for social media tools and analytics	Medium
Digital Asset Management	Organize and maintain organizational digital assets (images, videos, documents)	Medium

4.3 The Hybrid Support Model

For an organization of Yared Tour & Travel's size, a hybrid support model is recommended. This model combines:

Proactive Scheduled Visits (Weekly during stabilization, monthly/quarterly ongoing)

- Physical hardware inspection and maintenance
- Security patch management
- Backup verification
- Network health checks
- Printer and CCTV maintenance
- Staff feedback and issue identification

Reactive On-Demand Support (As needed)

- Remote resolution of software issues
- On-site dispatch for hardware failures
- Priority-based response (Critical: 4-hour on-site response)
- Issue logging and asset inventory updates

4.4 Required Resources for Effective ICT Management

To fulfill these responsibilities, the following resources are required:

Resource Category	Required Components	Estimated Cost
Spare Hardware Inventory	RAM modules, SSDs, network cables, spare Wi-Fi AP	
Software Repository	OS installers, Office suite, Finance software, security tools	
Diagnostic Tools	Bootable USB drives, network tester, cleaning supplies	
Technical Documentation	Network topology map, admin credentials, vendor contacts	
Financial Contingency	Annual budget for minor break-fix purchases	

5. Conclusions and Recommendations

5.1 Conclusions

The assessment of Yared Tour & Travel's ICT infrastructure and system support reveals an organization's interest to improve its technology utilizations. The existing technology environment, while functional in some areas, is characterized by fairly functioning hardware, inconsistent software, unreliable network performance. However, the changing tour and travel organization demands efficient Technology to meet the customers needs and stay also in the competitive sector. Hence, it vital to apply the recommendations as proposed by the IT professional

The organization's website and social media engagements, while visually appealing and mission-aligned, fail to function as an effective sales and marketing tool. The absence of customer based trip planning, bookable tours, detailed itineraries, and a booking or inquiry system creates a significant barrier between visitor interest and customer conversion. The website's broken navigation and missing essential pages (privacy policy, terms, FAQ) further undermine its effectiveness.

Social media presence, while established, is inconsistent and significantly underutilized. Given that most travelers post about their experiences on social media and that platforms like Instagram drive results for travel operators, this represents a substantial missed opportunity for brand awareness, engagement, and lead generation.

5.2 Recommendations

The following recommendations are organized into phased priorities to guide immediate action and long-term strategic development.

Phase 1: Immediate Stabilization (0-3 Months)

(Priority: High, Cost: Low-Medium)

- Action 1: Repair or replace PCs immediately to continue business operations.
- Action 2: Upgrade Desktops: Invest in RAM and SSD upgrades for the 3-4 most critical office PCs to improve performance at a low cost.
- Action 3: Standardize Software: Consolidate to a single office suite (Microsoft Office 365 recommended) and ensure all browsers are updated to the latest version.
- Action 4: Network Improvement: Implement a dedicated business-grade Wi-Fi solution and a separate network printer server to improve reliability.

Phase 2: System Upgrades & Optimization (3-12 Months)

(Priority: Medium, Cost: Medium-High)

- Action 5: Hardware Refresh: Begin a phased replacement of the oldest hardware (Desktops/Laptops) with new devices that have SSDs and 16GB+ RAM.
- Action 6: Windows 11 Migration: Develop a plan to upgrade or replace all Windows 10 devices before the October 2025 deadline.
- Action 7: Social media and Website & Service Review:
 - Build well structured and documented social media platform
 - Build a fully functional website which enhance trout and travel services
 - Audit the current website hosting and security (SSL, Plugins).
 - Implement a password manager for staff.
 - Review and potentially upgrade to a cloud-based solution.
- Action 8: Formalize IT Support: a dedicated IT Support Technician or outsource a managed IT service provider.

Phase 3: Strategic Alignment (12-24 Months)

(Priority: Low, Cost: High)

- Action 9: Cloud Migration: Begin migrating data and applications to the cloud (e.g., Microsoft 365, Google Workspace) to enable seamless remote work and data recovery.
- Action 10: Cybersecurity Audit: Conduct a full security audit to establish robust policies and defenses.
- Action 11: Digital Integration: Explore integrating the website, CRM, and Finance systems to automate manual tasks and provide a unified view of the client journey.

This investment should be compared against the cost of inaction: lost revenue from website conversion failures, productivity losses from aging hardware, potential data loss from inadequate backup systems, and reputational damage from operational failures.

5.4 Next Steps

1. Management Approval: Present this assessment to management for approval of the phased recommendations and associated budget.
2. Immediate Action (Week 1): Authorize repair or replacement of the Finance PC and initiate website fixes (broken navigation, CTA buttons, inquiry form).
3. Vendor Engagement (Weeks 2–4): Contact local IT service providers for quotes on hardware upgrades, network infrastructure, and managed service contracts.
4. IT Support Structure (Month 1): Make a decision on internal hire vs. outsourced MSP; initiate procurement process.
5. Phase 1 Kick-off (Month 1–2): Begin immediate stabilization activities including hardware upgrades, software standardization, and basic website improvements.
6. Progress Review (Month 3): Conduct a formal review of Phase 1 activities and adjust Phase 2 planning as needed.

This assessment provides a comprehensive roadmap for transforming Yared Tour & Travel's ICT infrastructure from a source of operational risk into a strategic enabler of sustainable growth and exceptional customer experience.